# A literature comparison matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for A literature comparison matrix before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Replication and reproduction

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Finding primary research

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

WORKBOOK · 2 OF 15

## Practise with fading help

> Complete the supported A literature comparison matrix trace and record the first step you can explain without help.

Use W2: Trace the worked case for A literature comparison matrix. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

## Individual reasoning check

> Check the proposed A literature comparison matrix explanation against the supplied facts

before discussing it.

Use the worked case for A literature comparison matrix. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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WORKBOOK · 4 OF 15

## Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Apply a new case independently

A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied A literature comparison matrix facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

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WORKBOOK · 6 OF 15

## W2 Trace the worked case

> Trace the worked case. Apply the supplied A literature comparison matrix facts and keep your reasoning.

Study A reports accuracy90% on easy classification; B reports claim support**80%** on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

WORKBOOK · 7 OF 15

## W3 Audit the limit

> Audit the limit. Apply the supplied A literature comparison matrix facts and keep your reasoning.

A report omits this qualification: A matrix documents comparison limits; it does not create missing common **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

WORKBOOK · 8 OF 15

## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied A literature comparison matrix facts and keep your reasoning.

A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

**My reasoning and evidence**

WORKBOOK · 9 OF 15

## W5 Design a revealing follow-up

Design a revealing follow-up. Apply the supplied A literature comparison matrix facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect A literature comparison matrix to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled a literature comparison matrix **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of a literature comparison matrix.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

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## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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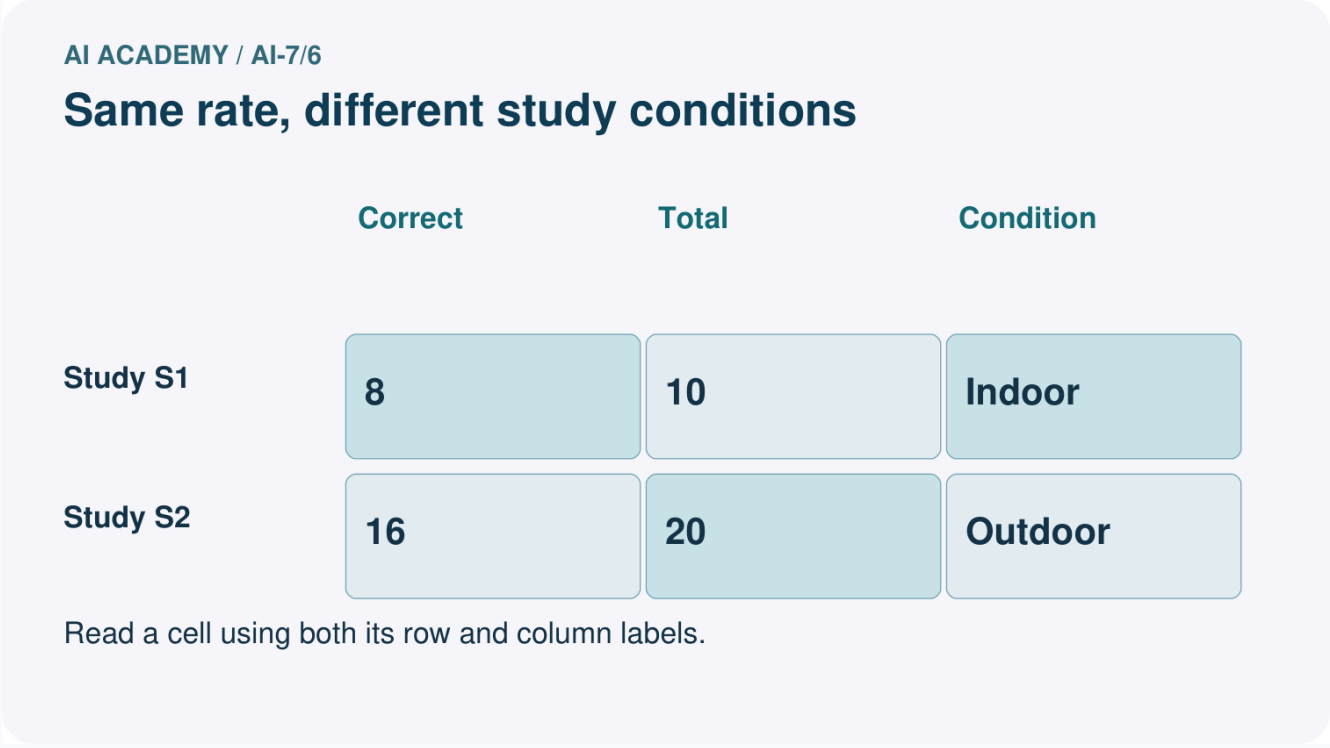
WORKBOOK · 12 OF 15

## Pictorial case practice: A literature comparison matrix

> A literature matrix that preserves differences. Two fictional studies evaluate related tools under different conditions. Explain the labelled step before checking the changed condition.

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.

Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/6

# A literature comparison matrix

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Record each method

Keep population, task, model version and scoring in separate columns.



## 2. Compare rates carefully

Both rates are 80%, with different denominators and conditions.



## 3. Name missing comparison

Neither study directly compares both tools on the same photos.

**CHANGE ONE THING**

Remove the indoor/outdoor column.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 14 OF 15

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Neither study directly compares both tools on the same photos.

1. Name missing comparison

2. Record each method

3. Compare rates carefully

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Name missing comparison Neither study directly compares both tools on the same photos. A **literature matrix** helps compare **evidence** without flattening differences.

Equal percentages do not mean equivalent tasks, samples or methods.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Remove the indoor/outdoor column. Which conclusion is supported by the supplied facts?

1. The matrix conceals an important condition difference.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The matrix conceals an important condition difference. A reader might now mistake unlike studies for a controlled head-to-head comparison.

### Repair a missing building block

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

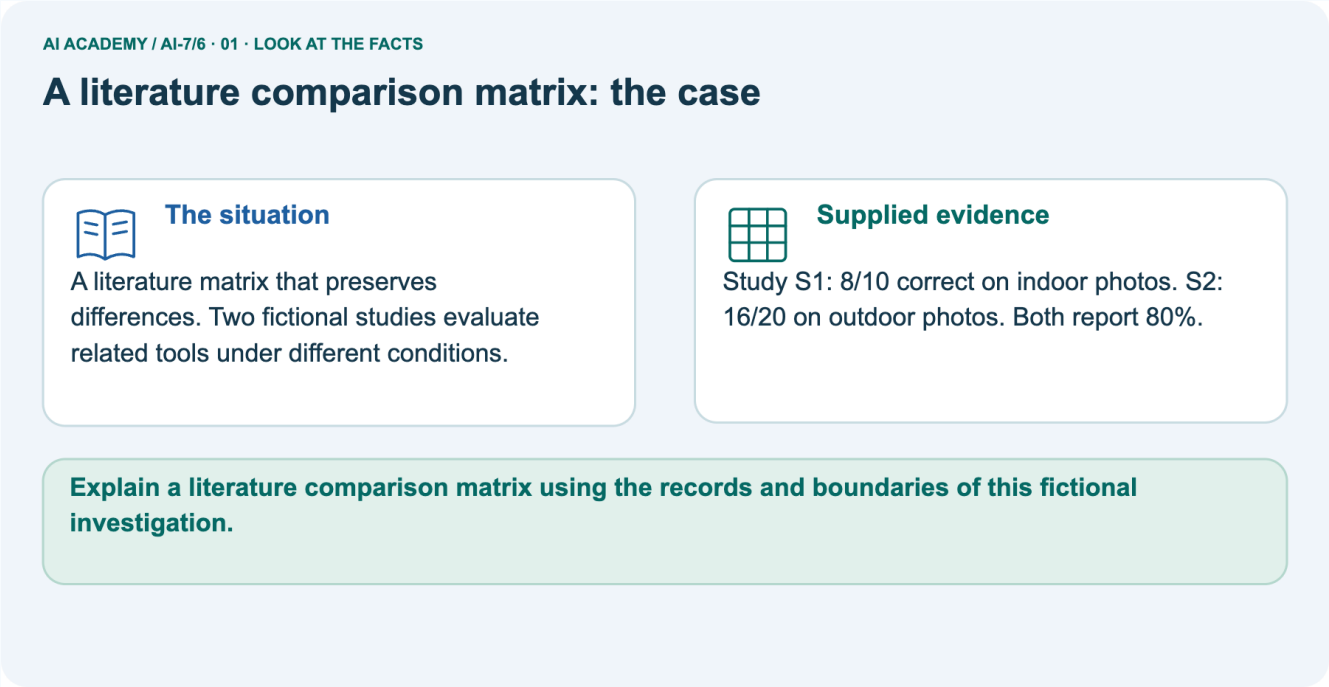
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

## Visual concept journey

> Explain each A literature comparison matrix picture from its labels, then record a relevant limit.

### 01. A literature comparison matrix: the case



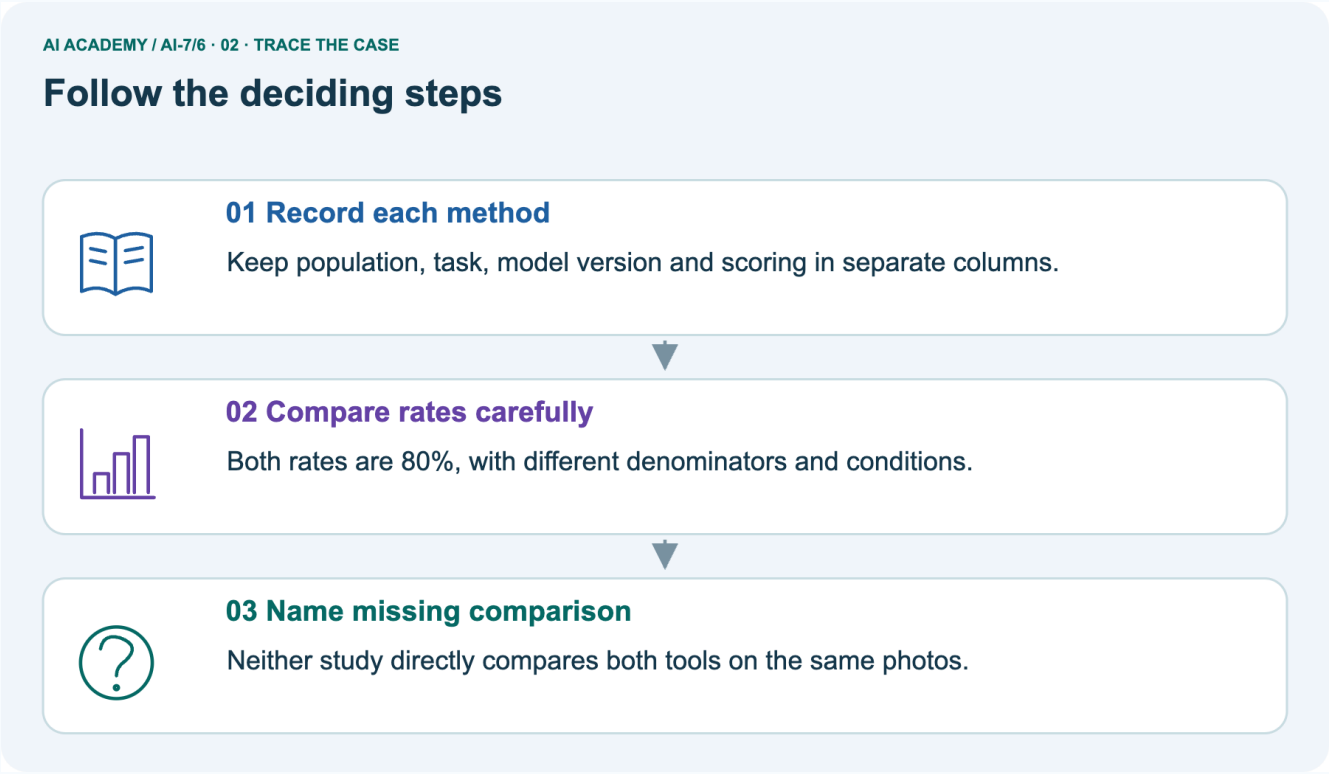

Follow the labelled picture with the matching explanation. Open the vector diagram

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions. Supplied facts: Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



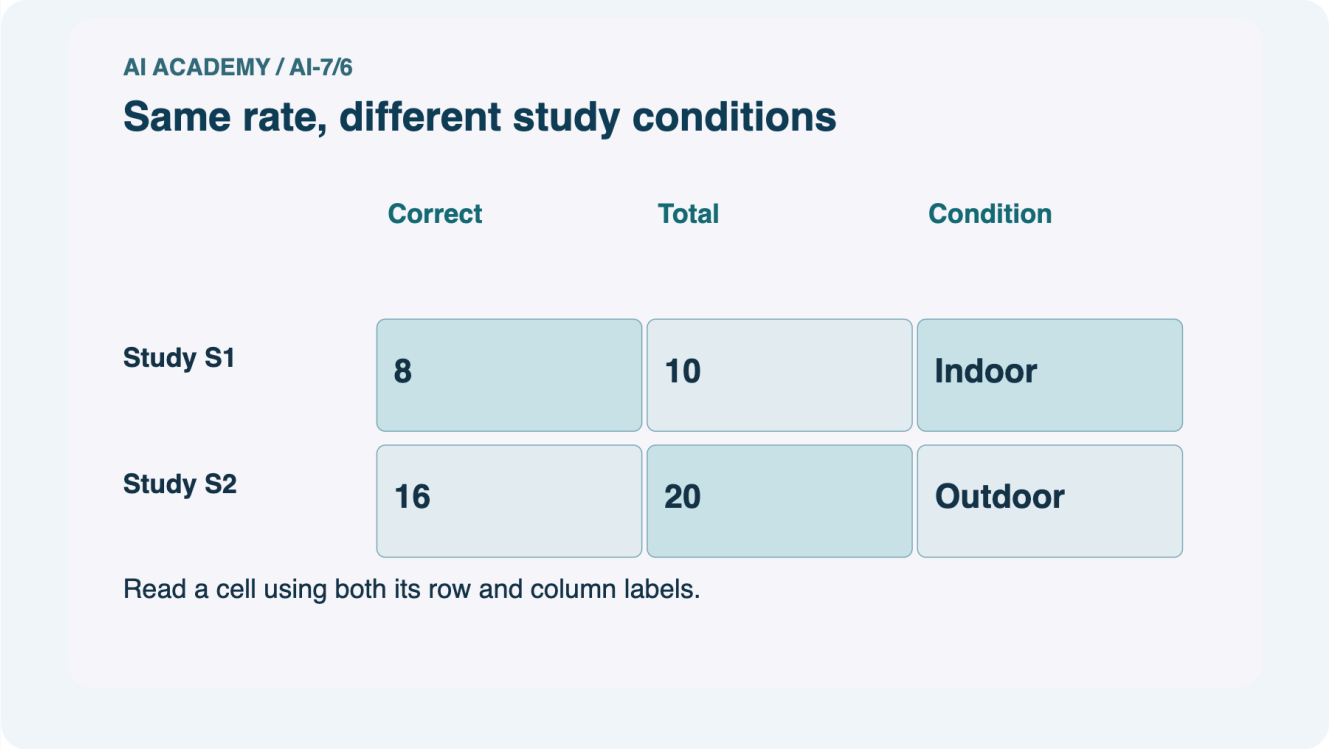

Follow the labelled picture with the matching explanation. Open the vector diagram

Record each method: Keep population, task, model version and scoring in separate columns. → Compare rates carefully: Both rates are **80%**, with different denominators and conditions. → Name missing comparison: Neither study directly compares both tools on the same photos.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

## 03. Same rate, different study conditions



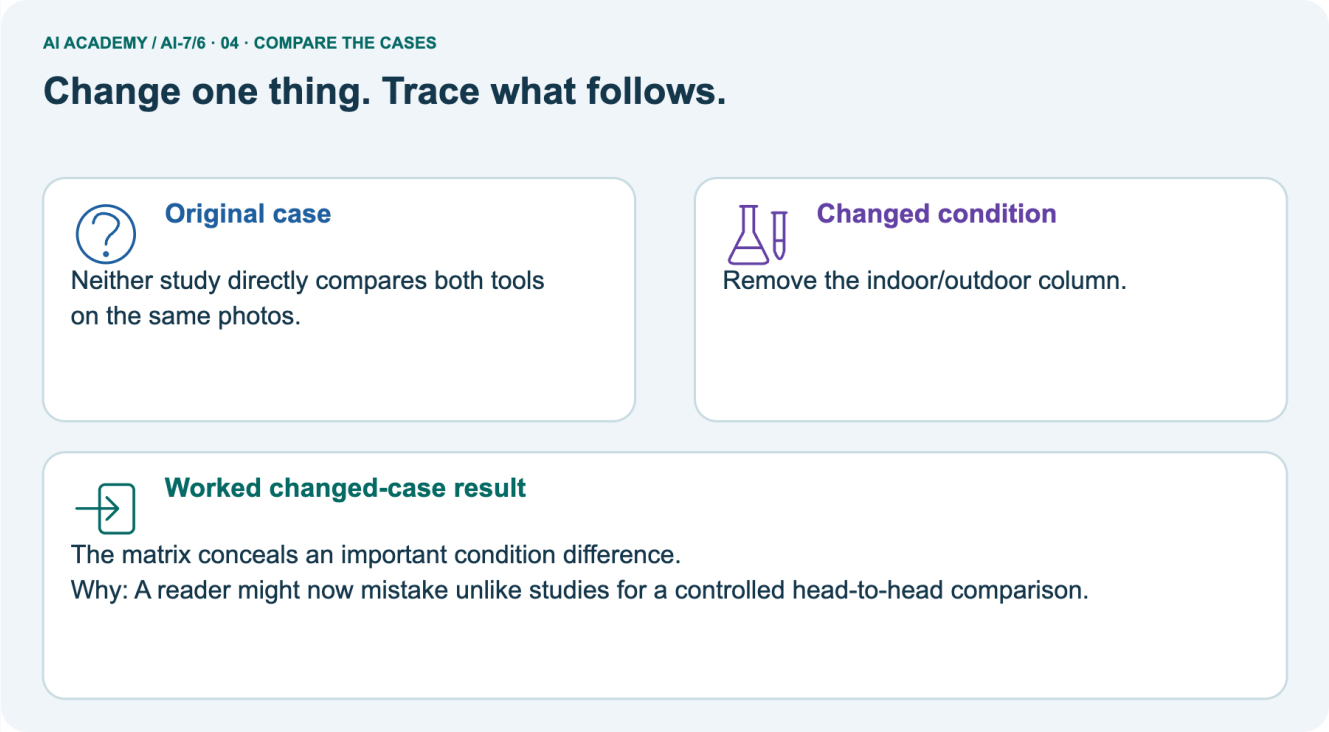

Follow the labelled picture with the matching explanation. Open the vector diagram

Same rate, different study conditions. Study S1: Correct = 8, Total = 10, Condition = Indoor. Study S2: Correct = 16, Total = 20, Condition = Outdoor.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

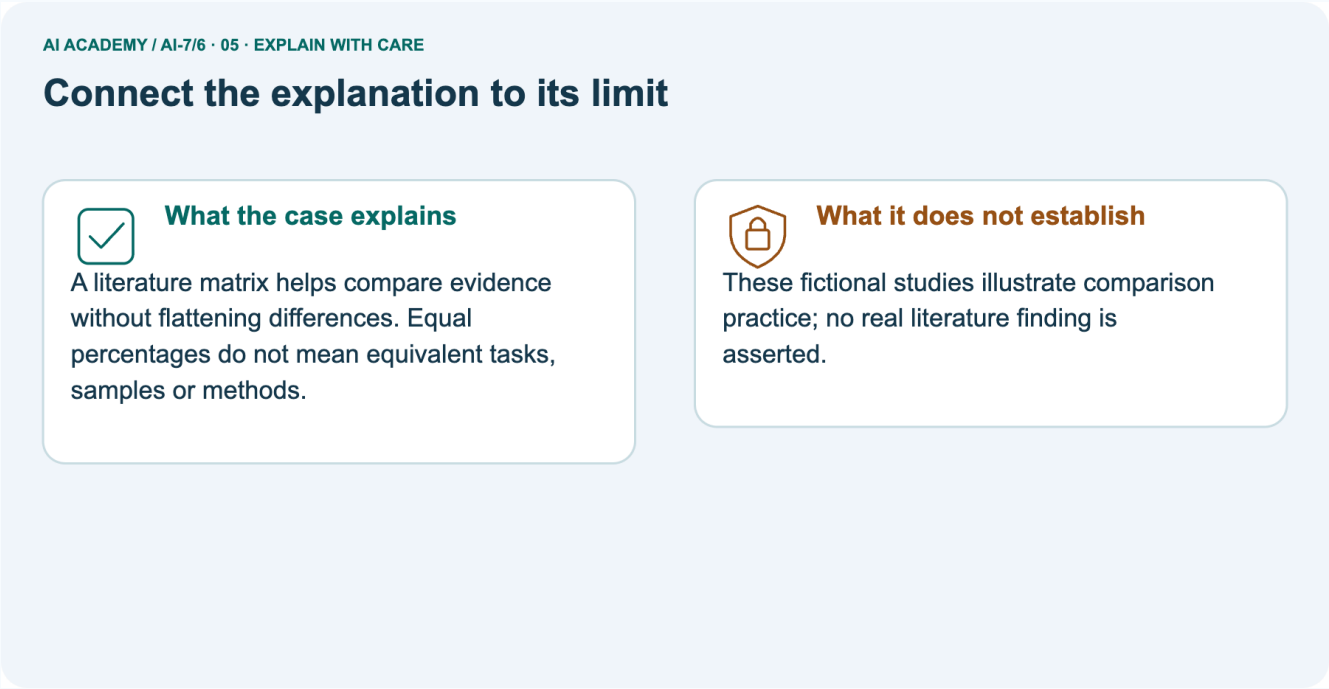
Original case: Neither study directly compares both tools on the same photos. Changed condition: Remove the indoor/outdoor column. Expected result: The matrix conceals an important condition difference.

Reason: A reader might now mistake unlike studies for a controlled head-to-head comparison.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

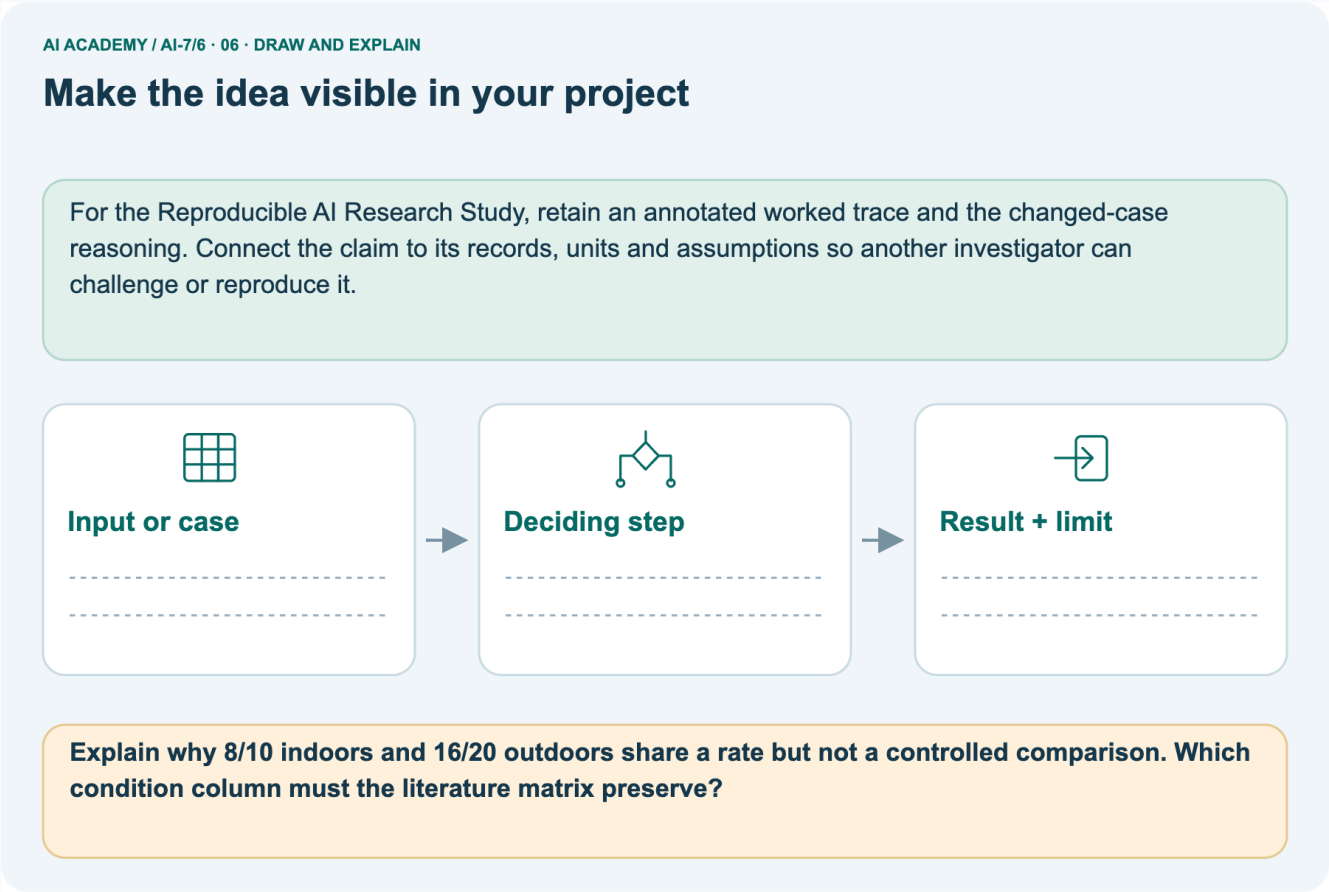
A **literature matrix** helps compare **evidence** without flattening differences. Equal percentages do not mean equivalent tasks, samples or methods.

Limit: These fictional studies illustrate comparison practice; no real literature finding is asserted.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.